

[EECS 304] Control I

Sheet (7) Control Design and Compensation

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Fall 2026

Outline

1 Overview

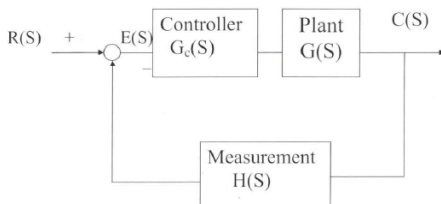
- Introduction
- P Controller
- PI Controller
- PD Controller
- PID Controller
- Summary

2 Problems

- Problem 1
- Problem 2
- Problem 3
- Problem 4
- Problem 5

Introduction

- What have we done so far?
 - 1 System Modelling
 - 2 Steady State Analysis
 - 3 Transient Response
 - 4 Stability
- What if the system's specifications need to be improved?
 - ▶ Compensation

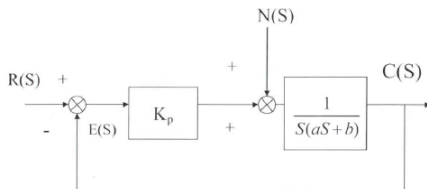


- **Goal:** Improving the specifications of the system (Steady State - Transient - Both)

P Controller

P Controller

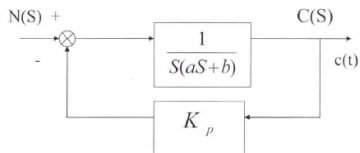
$$G_c(s) = K_p$$



• Specifications:

- ▶ Improve e_{ss} to a certain value.
- ▶ Decrease the damping in the system (the system is more oscillatory)
- ▶ As $K_p \uparrow\uparrow$, $e_{ss} \downarrow\downarrow$, $\zeta \downarrow\downarrow$

P Controller



$$\frac{C(s)}{N(s)} = \frac{1}{as^2 + bs + K_p}$$

$$\text{let } N(s) = \frac{1}{s}$$

$$\therefore c_{ss} = \lim_{s \rightarrow 0} s \frac{C(s)}{N(s)} * N(s) = \frac{1}{K_p}$$

Note

$$E(s) = R(s) - C(s) \rightarrow e_{ss} = r_{ss} - c_{ss}$$

$$\therefore e_{ss} = -c_{ss}$$

P Controller

∴ c/ch equation:

$$s^2 + \frac{b}{a}s + \frac{K_p}{a} = 0$$

$$\therefore s^2 + 2\zeta\omega_n s + \omega_n^2 = 0$$

$$\therefore \omega_n = \sqrt{\frac{K_p}{a}}, \quad 2\zeta\sqrt{\frac{K_p}{a}} = \frac{b}{a}$$

$$\therefore \zeta \propto \frac{1}{\sqrt{K_p}}$$

PI Controller

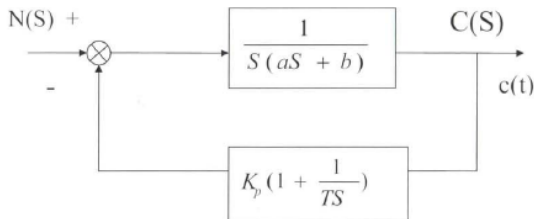
PI Controller

$$G_c(s) = K_p + \frac{K_I}{s} = K_p \left[1 + \frac{1}{T_I s} \right]$$

- **Specifications:**

- ▶ Eliminate the steady state error.
- ▶ Increase the settling time of the system.

PI Controller



$$\frac{C(s)}{N(s)} = \frac{s}{as^3 + bs^2 + K_p s + \frac{K_p}{T}}$$

$$\therefore c_{ss} = \lim_{s \rightarrow 0} s \frac{C(s)}{N(s)} * N(s) = 0$$

Note

If I used I without K_p , the system will be unstable (Try it!)

PD Controller

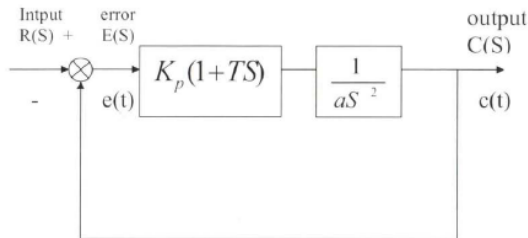
PD Controller

$$G_c(s) = K_p + K_d s = K_p[1 + T_d s]$$

- **Specifications:**

- ▶ Used to achieve **transient response specifications** or a **desired location of the dominant poles**.
- ▶ Improve the transient response ($M_p \downarrow\downarrow$, $t_s \downarrow\downarrow$).
- ▶ Adds damping to the system ($\zeta \uparrow\uparrow$) and thus permits the use of larger value of the gain K , which will decrease e_{ss} .

PD Controller



$$\frac{C(s)}{R(s)} = \frac{K_p(1 + T_d s)}{s^2 + \frac{K_p}{a} T_d s + \frac{K_p}{a}}$$

$$\therefore \omega_n = \sqrt{\frac{K_p}{a}}, \quad \zeta = \frac{T_d}{2} \sqrt{\frac{K_p}{a}}$$

PID Controller

PID Controller

$$G_c(s) = K_p + K_d s + \frac{K_I}{s} = K_p \left[1 + T_d s + \frac{1}{T_I s} \right]$$

- **Specifications:**

- ▶ Used to achieve **transient response specifications** or **steady state specifications**.
- ▶ Improve the transient response ($M_p \downarrow\downarrow, t_s \downarrow\downarrow$) and eliminate the steady state error.

Summary

- **P:**
 - ▶ Decrease ss error.
 - ▶ Bad effect on transient.
- **PI:**
 - ▶ Eliminate ss error.
- **PD:**
 - ▶ Improve ss error.
 - ▶ Improve transient.
- **How to choose controller type:**
 - 1 $e_{ss}, K_p, K_v, K_a \rightarrow$ PI
 - 2 $\zeta, \omega_n, t_r, t_p, t_s, M_p$, dominant poles location $\rightarrow$ PD
 - 3 1 & 2 $\rightarrow$ PID

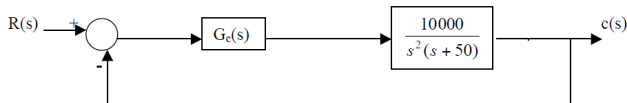
Problem 1

Problem 1

The block diagram of a type 2 system with a cascade controller $G_c(s)$ is shown in figure. The objective is to design a controller $G_c(s)$ so that:

- Maximum overshoot $< 10\%$
- Settling time < 1

Design the controller that satisfies the performance specifications given above.



Problem 1

Solution

We need an improvement in transient response $\rightarrow$ PD Controller

$$\therefore G_c(s) = K_p(1 + T_d s) \quad \text{OR} \quad G_c(s) = K_p + K_d s$$

$$\therefore M_p = 0.1 \rightarrow \zeta = 0.59115$$

$$\therefore t_s = \frac{4}{\zeta \omega_n} = 1 \rightarrow \omega_n = \frac{4}{1 * 0.59115} = 6.766 \text{ rad/sec}$$

c/ch Equation: $1 + G(s)H(s) = 0$

$$\therefore G(s) = G_c(s)G_p(s) = \frac{10000(K_p + K_d s)}{s^2(s + 50)}$$

$$\therefore T.F. = \frac{10000(K_p + K_d s)}{s^2(s + 50) + 10000(K_p + K_d s)}$$

c/ch Equation:

$$s^3 + 50s^2 + (10000K_d)s + 10000K_p = 0$$

Problem 1

Solution

$$\therefore s^3 + 50s^2 + (10000K_d)s + 10000K_p = 0$$

$$\therefore (s + \alpha)(s^2 + 2\zeta\omega_n s + \omega_n^2) = 0$$

$$\therefore s^3 + (2\zeta\omega_n + \alpha)s^2 + (\omega_n^2 + 2\zeta\omega_n\alpha)s + \alpha\omega_n^2 = 0$$

coeff. of s^2

$$50 = 2\zeta\omega_n + \alpha \rightarrow \alpha = 42 > 5\zeta\omega_n$$

coeff. of s

$$10000K_d = \omega_n^2 + 2\zeta\omega_n\alpha$$

coeff. of s^0

$$10000K_p = \alpha\omega_n^2 \rightarrow K_p = 0.19$$

$$\therefore K_d = 0.038175$$

$$\therefore G_c(s) = 0.19 + 0.038175s \quad \text{OR} \quad G_c(s) = 0.19(1 + 0.203s)$$

Problem 2

Problem 2

Consider a unity feedback controller system whose feedforward transfer function is:

$$G(s) = \frac{2}{s(s+2)(s+8)}$$

- Find the static error coefficients and the error as a function of time for unit ramp input.
- Design a compensator so that the closed loop dominant poles are located at $s = -1 \pm j2\sqrt{3}$

Problem 2

Solution

a)

$$K_p = \lim_{s \rightarrow 0} G(s)H(s) = \infty$$

$$K_v = \lim_{s \rightarrow 0} sG(s)H(s) = \frac{2}{16} = \frac{1}{8}$$

$$K_a = \lim_{s \rightarrow 0} s^2 G(s)H(s) = \frac{2}{16} = 0$$

For unit ramp $\rightarrow e_{ss} = \frac{1}{K_v} = 8$

b) We need an improvement in transient response $\rightarrow$ PD Controller

$$\therefore G_c(s) = K_p(1 + T_d s) \quad \text{OR} \quad G_c(s) = K_p + K_d s$$

Given the dominant poles $\rightarrow s = -1 \pm j2\sqrt{3}$

$$\therefore -1 \pm j2\sqrt{3} = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$$

Problem 2

Solution

$$\therefore -1 \pm j2\sqrt{3} = -\zeta\omega_n \pm j\omega_n\sqrt{1-\zeta^2}$$

$$\therefore -1 = -\zeta\omega_n, \quad 2\sqrt{3} = \omega_n\sqrt{1-\zeta^2}$$

$$\therefore \omega_n = \sqrt{13}, \quad \zeta = \frac{1}{\sqrt{13}}$$

using PD $\rightarrow G_c(s) = K_p(1 + T_d s)$

$$\therefore G(s) = G_c(s)G_p(s) = \frac{2K_p(1 + T_d s)}{s(s+2)(s+8)}$$

$$\therefore T.F. = \frac{2K_p(1 + T_d s)}{s(s+2)(s+8) + 2K_p(1 + T_d s)}$$

c/ch Equation:

$$s^3 + 10s^2 + (16 + 2K_p T_d)s + 2K_p = 0$$

Problem 2

Solution

c/ch equation:

$$s^3 + 10s^2 + (16 + 2K_p T_d)s + 2K_p = 0$$

$$s^3 + (2\zeta\omega_n + \alpha)s^2 + (\omega_n^2 + 2\zeta\omega_n\alpha)s + \alpha\omega_n^2 = 0$$

coeff. of s^2

$$2 + \alpha = 10$$

coeff. of s

$$2\alpha + 13 = 16 + 2K_p T_d$$

coeff. of s^0

$$13\alpha = 2K_p$$

$$\therefore K_p = 52, \quad \alpha = 8, \quad T_d = 0.125$$

$$\therefore G_c(s) = 52(1 + 0.125s)$$

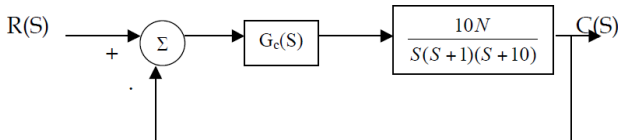
Problem 3

Problem 3

The following figure shows the block diagram of the liquid level control system. The liquid level is represented by $c(t)$, and N denotes the number of inlets. It is desired that $N=20$. Design a controller $G_c(S)$ such that:

- The overshoot is zero.
- The tank is filled to the reference level within 2.5 sec.

What is the effect of the controller on the steady state response? And why?



Problem 3

Solution

a) We need an improvement in transient response $\rightarrow$ PD Controller

$$\therefore G_c(s) = K_p(1 + T_d s)$$

$$\therefore M_p = 0 \rightarrow \zeta = 1$$

$$\therefore t_s = \frac{4}{\zeta \omega_n} = 2.5 \rightarrow \omega_n = \frac{4}{2.5} = 1.6 \text{ rad/sec}$$

c/ch equation:

$$s^3 + 11s^2 + (10 + 200K_p T_d)s + 200K_p = 0$$

$$s^3 + (2\zeta\omega_n + \alpha)s^2 + (\omega_n^2 + 2\zeta\omega_n\alpha)s + \alpha\omega_n^2 = 0$$

Problem 3

Solution

coeff. of s^2

$$2\zeta\omega_n + \alpha = 11$$

coeff. of s

$$\omega_n^2 + 2\zeta\omega_n\alpha = 10 + 200K_pT_d$$

coeff. of s^0

$$\alpha\omega_n^2 = 200K_p$$

$$\therefore K_p = 0.1, \quad \alpha = 7.8, \quad T_d = 0.876$$

$$\therefore G_c(s) = 0.1(1 + 0.876s)$$

Problem 3

Solution

b) $\therefore$ the system is Type 1 $\rightarrow$ Use ramp as input

$$e_{ssBefore} = \frac{1}{K_v} = \frac{1}{20}$$

$$e_{ssAfter} = \frac{1}{0.1 * 20} = \frac{1}{2}$$

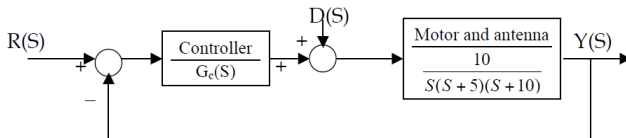
As shown, the steady state error increases as we decrease the steady state gain using small value of K_p

Problem 4

Problem 4

A large antenna is used to receive satellite signals and must accurately track the satellite as it moves across the sky. The control system uses an armature controlled motor and a controller to be selected as shown in figure. The system specifications require an overshoot to a step input less than or equal 10 % with a settling time less than or equal 2 seconds.

- Design a controller $G_c(S)$ to achieve the required specifications and plot the resulting response.
- Determine the effect of the disturbance $[D(S)=Q/S]$ on the steady state output.



Problem 4

Solution

a) We need an improvement in transient response $\rightarrow$ PD Controller

$$\therefore G_c(s) = K_p(1 + T_d s)$$

$$\therefore M_p = 0.1 \rightarrow \zeta = 0.59$$

$$\therefore t_s = \frac{4}{\zeta \omega_n} = 2 \rightarrow \omega_n = \frac{4}{2 * 0.59} = 3.38 \text{ rad/sec}$$

c/ch equation:

$$s^3 + 15s^2 + (50 + 10K_p T_d)s + 10K_p = 0$$

$$s^3 + (2\zeta\omega_n + \alpha)s^2 + (\omega_n^2 + 2\zeta\omega_n\alpha)s + \alpha\omega_n^2 = 0$$

Problem 4

Solution

coeff. of s^2

$$2\zeta\omega_n + \alpha = 15$$

coeff. of s

$$\omega_n^2 + 2\zeta\omega_n\alpha = 50 + 10K_p T_d$$

coeff. of s^0

$$\alpha\omega_n^2 = 10K_p$$

$$\therefore K_p = 12.6, \quad \alpha = 11, \quad T_d = 0.04323$$

$$\therefore G_c(s) = 12.6(1 + 0.04323s)$$

Problem 4

Solution

b)

$$\frac{Y(s)}{D(s)} = \frac{10}{s(s+5)(s+10) + 10K_p(1 + T_d s)}$$

$$\therefore y_{ss} = \lim_{s \rightarrow 0} sY(s) = \frac{10Q}{10K_p} = \frac{Q}{K_p}$$

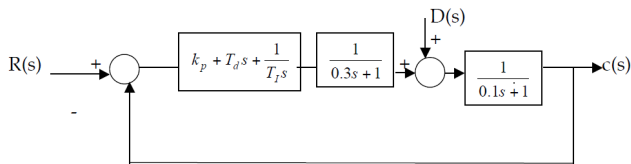
As shown, the steady state output decreases by factor K_p which is better.

Problem 5

Problem 5

A control system with a type 0 process and PID controller is shown in figure . Design the parameters of the PID controller so that the following specifications are satisfied:

- Settling time ≤ 0.2
- Max. Overshoot ≤ 5 percent.
- Steady state error due to unit step disturbance = zero.



Problem 5

Solution

$$\therefore M_p = 0.05 \rightarrow \zeta = 0.69$$

$$\therefore t_s = \frac{4}{\zeta \omega_n} = 0.2 \rightarrow \omega_n = \frac{4}{0.2 * 0.69} = 28.98 \text{ rad/sec}$$

$$\frac{C(s)}{R(s)} = \frac{K_p + T_d s + \frac{1}{T_i s}}{(0.1s + 1)(0.3s + 1) + (K_p + T_d s + \frac{1}{T_i s})}$$

c/ch equation:

$$s^3 + \frac{1}{0.03}(0.4 + T_d)s^2 + \frac{1 + K_p}{0.03}s + \frac{1}{0.03 T_i} = 0$$

$$s^3 + (2\zeta\omega_n + \alpha)s^2 + (\omega_n^2 + 2\zeta\omega_n\alpha)s + \alpha\omega_n^2 = 0$$

Problem 5

Solution

coeff. of s^2

$$2\zeta\omega_n + \alpha = \frac{1}{0.03}(0.4 + T_d)$$

coeff. of s

$$\omega_n^2 + 2\zeta\omega_n\alpha = \frac{1 + K_p}{0.03}$$

coeff. of s^0

$$\alpha\omega_n^2 = \frac{1}{0.03T_i}$$

$\therefore$ I have 4 unknowns in 3 equations $\rightarrow \alpha = 5\zeta\omega_n = 100$

$$\therefore K_p = 144.2, \quad T_i = 3.9 * 10^{-4}, \quad T_d = 3.8$$

$$\therefore G_c(s) = 144.2 + 3.8s + \frac{1}{3.9 * 10^{-4}s}$$

Thank you!

Questions?

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